

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Master of Computer Applications (MCA)

Semester – II University Examination May 2010

MA702 Computer Oriented Numerical and Statistical Techniques

Date: 21/05/2010, Friday

Time: 12:30 noon to 3:30 p.m.

Maximum Marks: 70

Instructions:

1. Use separate answer book for each section
2. Figure to the right indicates full marks of the respective question
3. Use of scientific (non – programmable) calculator is permitted
4. Statistical tables will be provided at exam

SECTION I

Q. 1. (a) Answer the following questions.

06

1. A hypothetical computer stores floating point base 10 numbers with three decimal digits in mantissa, one decimal digit for exponent. Show the results of following arithmetic operations performed on computer. Assume rounding of results.

i $1.37 + 0.0269 = 0.140 \times 10^0$
ii $4850 - 4820 = 0.300 \times 10^2$
iii $3780 - 0.321 = 0.378 \times 10^4$

2. The decimal representation of binary number 1.0110101 is 1.6140525
3. The number $\hat{a}$ is said to approximate a to r significant digits if r is the largest nonnegative integer for which the relative error R , satisfies the condition,

(i) $R < \frac{10^{1-r}}{2}$ (ii) $R < \frac{2}{10^{1-r}}$ (iii) $|R| < \frac{10^{1-r}}{2}$ (iv) $|R| < \frac{2}{10^{1-r}}$

4. For finding the real root of nonlinear algebraic equation $f(x)=0$ exists in the interval $[a,b]$, the bisection method converges in n iterations if the absolute error A satisfies the condition $A \leq \frac{1}{2^n}(b-a)$. (True / False)

5. For finding the real root of nonlinear algebraic equation $f(x)=0$ exists in the interval $[a,b]$, the false position method depends on both sign change of $f(x)$ and values of $f(x)$. (True / False)

6. The Newton Raphson method to find real root of nonlinear algebraic equation $f(x)=0$ may converges when the initial guess x_0 satisfies the condition $|f'(x_0)| > 0$. (True / False)

- a. Discuss various types of errors that may occur in digital computer.

03

Q. 2. (a) Use method of false position to find the real root of the equation $f(x)=0$, where $f(x)=x^3+x^2-3x-3$, correct up to three significant digits.

04

(b) Use Newton Raphson method to find the real root of the equation $f(x)=0$ where $f(x)=3x+\sin x-e^x$, correct up to three significant digits.

05

- (c) Discuss the algorithm of Bisection method to find the real root of the equation $f(x)=0$ 04

OR

- Q.2. (a) Use method of fixed point iteration to find the real root of the equation $f(x)=0$, where $f(x)=x^2+x-1$, correct up to three significant digits. 05
- (b) Use method of bisection to find the real root of the equation $f(x)=0$, where $f(x)=x^3-2x+1$, correct up to three significant digits. 04
- (c) What do you mean by the order of convergence of iterative method for finding real root of algebraic equation $f(x)=0$. Give the order of convergence of Bisection, False position and Newton Raphson methods 04

- Q.3. (a) Use Gauss elimination method to find the solution of simultaneous linear 04

$$\begin{aligned} 4x_1 - 2x_2 + x_3 &= 15 \\ -3x_1 - x_2 + 4x_3 &= 8 \\ x_1 - x_2 + 3x_3 &= 13 \end{aligned}$$

- (b) Use Cholesky factorization method to find the solution of simultaneous 05

$$\begin{aligned} 3x_1 - x_2 + 2x_3 &= 12 \\ x_1 + 2x_2 + 3x_3 &= 11 \\ 2x_1 - 2x_2 - x_3 &= 2 \end{aligned}$$

- (c) What do you mean by interpolation? Use appropriate interpolation formula that passes through each points 04

x	-1.1	2.1	3.5
y = f(x)	1.3	2.5	-1.6

Use this formula to interpolate $f(2.4)$

OR

- Q.3. (a) Use Gauss Jacobi method to find the solution of simultaneous linear 05

$$\begin{aligned} x_1 + 2x_2 - 5x_3 &= -1 \\ 6x_1 - 2x_2 + x_3 &= 11 \\ -2x_1 + 7x_2 + 2x_3 &= 5 \end{aligned}$$

Perform four iterations only.

- (b) Use a Newton -Gregory interpolating polynomial of degree 3 from the table for four points at $x = 0.4$ to $x = 1.0$ 04

x	0.0	0.2	0.4	0.6	0.8	1.0	1.2
y = f(x)	0.000	0.203	0.423	0.684	1.030	1.557	2.572

Use this polynomial to interpolate $f(0.53)$

- (c) The following table has values for $f(x)$ 04

x	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8
f(x)	1.543	1.669	1.811	1.971	2.151	2.352	2.577	2.828	3.107

Integrate $f(x)$ between $x = 1.0$ to $x = 1.8$ using Simpson's $\frac{1}{3}$ rule.

SECTION II

- Q. 4.(a) Define the following terms with suitable examples to each of them 06
- i Sample space
 - ii Mathematical definition of probability
 - iii Mutually exclusive events
- (b) Answer the following questions 03
1. If A and B are mutually exclusive events, $P(A) = 0.29$, and $P(B) = 0.43$, then $P(A \cap B^c) =$ _____
 2. If $P(A) = 0.40$, $P(B) = 0.30$, and $P(A \cap B) = 0.15$ then $P(A|B^c) =$ _____
 3. If $P(A) = 0.8$, $P(B) = 0.5$ and $P(A \cup B) = 0.9$, then A and B are independent events. (True / False)
- Q.5. (a) A die is rolled twice. Describe the sample space of elementary events. 04
Describe the events A: the sum of the points shown equal 8 and B: a 6 will appear at least once. Find the probabilities of events A and B under the condition that all elementary events are equally likely.
- (b) Suppose that in a particular city, airport A handles 50% of all airline traffic, and airports B and C handle 30% and 20% respectively. The probabilities of detecting weapons at the three airports are 0.9, 0.8 and 0.85. If a passenger at one of the airports is found to be carrying a weapon through boarding gate, what is the probability that the passenger is using airport A? 05
- (c) A very large batch of components has arrived at a distributor. The batch can be characterized as acceptable only if the proportion of defective component is 0.1. The distributor decides to randomly select 10 components and to accept the batch only if the number of defective components in the sample is atmost 2. What is the probability that the batch will be accepted? 04
- OR**
- Q.5. (a) The computers of six faculty members in a certain department are to be replaced. Two of the faculty members have selected laptop machines and the other four have chosen desktop machines. Two computers are selected at random from six computers for setup on particular day, what is the probability that both laptop machines are selected for setup? 04
- (b) At a certain gas station, 40% of the customers use regular gas, 35% use premium gas and 25% use extra premium gas. Of those customers using regular gas only 30% fill their tanks. Of those customers using premium gas 60% fill their tanks, where as of those customers using extra premium, 50% fill their tanks. What is the probability that the next customer will request premium gas and fill the tank? 04
- (c) A telephone switch board receives 12 calls on an average during an hour. Find the probability that during a period of 10 minutes,
a. No calls received. b. Exactly 3 calls received. c. More than 2 calls received. 05

Q.6. (a) The mean and standard deviation of marks of 500 students in an examination are 54 and 6 respectively. If the distribution of the marks is approximately normal, Find the probability that

1. Student gets marks more than 60.
2. Student gets marks between 48 and 56 and
3. If the standard of passing is 38 marks find the number of students failing in the examination

(b) The medical insurance company's records indicate that the average length of stay at a hospital for a patient after open-heart surgery is 10 days. A medical research group wants to reconfirm this average stay. A random sample of 36 such patients was taken from the record files and it was determined that the sample mean was 8.5 days with a standard deviation of 3.3 days. Using 5% significance level check if the insurance company's records are justified. 04

(c) Discuss the types of errors in statistical hypothesis testing. 04

OR

Q.6. (a) The manager of a department store is thinking about establishing a new billing system for the store's credit customers. After a thorough financial analysis, she determines that the new system will be cost – effective only if the mean monthly account is more than \$170. A random sample of 400 monthly accounts is drawn, for which the sample mean is \$178. The manager knows that the accounts are approximately normally distributed with a standard deviation of \$65. Can the manager conclude from this that the new system will be cost – effective? Use 5% level of significance. 04

(b) The height of adult males is normally distributed with a mean of 172 cm and standard deviation of 8 cm. If 90% of adult males exceed a certain height, Calculate this height. 04

(c) The Survey of Study Habits and Attitudes (SSHA) is a psychological test that evaluates college students' motivation, study habits, and attitudes toward school. A private college gives the SSHA to 10 of its incoming first-year women students. Their scores are 05

154	109	137	115	152	154	178	101	103	126
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Compute the mean and standard deviation of SSHA score.
